Paresh Prajapati

AI/ML Engineer | Generative AI Specialist

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SUMMARY

AI/ML Engineer with 3+ years of experience taking Generative AI products from design to production. Shipped three live AI platforms — a catering marketplace assistant, a compliance and safeguarding engine for schools, and a travel companion — alongside agentic report writers, text-to-SQL agents with strict data guardrails, and a fine-tuning data pipeline. Builds on LangChain/LangGraph, hybrid RAG, and async FastAPI services on AWS, with a focus on evaluation, human oversight, cost efficiency, and latency.

TECHNICAL SKILLS

GenAI & Agents: LLMs, LangChain, LangGraph, Deep Agents, ReAct & tool-calling agents, multi-agent workflows, text-to-SQL, prompt engineering, structured outputs (Pydantic)

RAG & Search: Hybrid dense + sparse retrieval, Qdrant, OpenSearch, FastEmbed, Docling, chunking & embedding pipelines, metadata-filtered retrieval

LLM Providers: AWS Bedrock, Anthropic Claude, OpenAI, OpenRouter

Speech, Vision & Fine-tuning: AssemblyAI (speaker diarization), AWS Textract, Tesseract OCR, QLoRA, synthetic training data, Hugging Face Transformers

Backend: Python, FastAPI, Django, Flask, REST APIs, SSE streaming, ARQ/Redis job queues, JWT auth

Data: PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Alembic, sqlglot, PrestoDB, Cassandra

Cloud & DevOps: AWS (ECS Fargate, S3, Bedrock, Textract), Docker, Docker Compose

LLMOps & Quality: LLM evals with labelled sets & replay, OpenTelemetry tracing (Arize Phoenix), Sentry, token & cost tracking, pytest, mypy --strict, ruff, pre-commit

ML Libraries: Pandas, NumPy, TensorFlow, Scikit-learn

EXPERIENCE

AI Engineer — Encodedots

Feb 2026 – Present

- Own end-to-end AI solution development — system design, agent architecture, deployment, and monitoring on AWS.
- Design scalable multi-agent workflows (LangGraph) powering conversational assistants, semantic search, and document intelligence in production.
- Drive cost efficiency across LLM workloads through model selection, caching strategies, token-usage tracking, and streaming architectures.

Software Engineer (AI/ML)

Feb 2023 – Oct 2025

- Built and shipped AI/ML solutions optimizing operational processes — RAG frameworks, LLM integrations, and NLP pipelines.
- Collaborated across all project stages, from problem definition and data preparation to model evaluation and delivery.
- Developed production chatbots, document-analysis systems, and recommendation engines used by real clients.

FEATURED PRODUCTION PROJECTS

CaterNest — AI-Powered Catering Marketplace

In Production

- **Business:** built the AI layer of a catering marketplace — customers assemble orders conversationally in plain English, and caterers digitize PDF/photo menus into structured, searchable listings in minutes instead of hours of manual entry.
- **Agent stack:** LangGraph multi-step agent on AWS Bedrock (Claude) with tool-calling into hybrid keyword + vector semantic search over OpenSearch.
- **Production hardening:** persistent conversation memory via LangGraph PostgreSQL checkpointer, PII-safe SSE token streaming, Redis caching, containerized services on ECS Fargate.

- **Menu extraction:** AWS Textract OCR + LLM structuring pipeline converting raw menu documents into normalized catalog data.

PolicyGuard — AI Compliance & Safeguarding Engine for Schools

In Production

- **Business:** AI engine for a school safeguarding and compliance platform — policy scans against the legislation register with gap lists and rewrites, incident and complaint triage, excursion risk assessments, supplier certificate checks, CV parsing, and an AI tool safety scanner graded against AI-in-education frameworks.
- **Async job engine:** FastAPI with ARQ/Redis workers; every service queues work, returns a job ID immediately, and streams staged progress over SSE with cancel and retry. Contract-first Pydantic JSON lets the backend switch from mock to live with a config flag.
- **Hybrid RAG:** dense + sparse retrieval on Qdrant (FastEmbed) over legislation and framework libraries, scoped by jurisdiction and category; Docling parses PDF/DOCX for ingestion, with stale chunks evicted when a document is replaced.
- **Safety & quality:** escalate-on-doubt triage with human sign-off (AI can raise severity, never lower it); every scan records prompt version, model, and engine build; an eval harness replays labelled cases; per-request cost tracking via OpenRouter.

IsleWay — AI Travel Companion

In Production

- **Business:** AI concierge for an island travel platform — answers destination, itinerary, and logistics questions from curated knowledge, turning static content into an interactive planning experience.
- **Backend:** FastAPI service with clean route → controller → service architecture, agentic chat with knowledge-base retrieval, and background workers.
- **Production hygiene:** per-request tracing, structured logging, uniform response envelopes, centralized error handling, strict typing (mypy --strict), Docker; consumed by a Flutter cross-platform client.

ADDITIONAL PROJECTS

- **FinScribe:** agent that writes and edits financial reports — a LangChain ReAct agent (Deep Agents) behind FastAPI, streamed over SSE. Every figure is bound to the calculation engine instead of typed by the model, so the agent cannot invent a number. LangGraph checkpointing on Redis plus a custom MySQL saver, OpenTelemetry tracing (Arize Phoenix), token cost tracking.
- **CarePilot:** role-based assistant for a care-services app with per-role toolsets (participant, carer, support worker). Text-to-SQL runs through a guard that parses each query (sqlglot), allows reads only, and rewrites it to the user's permitted data. Also generates on-demand AI reports; FastAPI + ARQ workers on ECS.
- **AthleteIQ:** generates baseline, concussion, and injury assessment reports from athlete data with LangChain + Claude, and treatment plans from session recordings via AssemblyAI speaker diarization; async Redis job queues, MySQL, JWT-secured API.
- **MenuForge:** training-data pipeline for an OCR → structured JSON fine-tune. Labels are generated first, then typeset into menu pages and degraded like real photos, so ground truth is exact by construction; every page is verified against its label. Set up for QLoRA fine-tuning of a vision-language model.
- **QueryMind:** LLM converts natural-language questions into SQL executed via PrestoDB over Cassandra, opening complex datasets to non-technical users.
- **ClauseLens:** RAG framework for legal document analysis — precise extraction with citations, in-document highlighting, and clause comparison across contracts.
- **AskHR:** voice-enabled agentic assistant with dynamic tool triggering for leave balances, holiday lookups, and RAG-grounded policy answers.
- **MatchMind:** LLM-based matchmaking recommendations with demographic rules; early-stopping cut unnecessary API spend and parallel batch execution processed thousands of profiles efficiently.

EDUCATION

B.Tech, Information & Communication Technology (ICT) — Sankalchand Patel University, Visnagar | CGPA: 8.33 — 2019 – 2023

Class XII — Gurkul School of Science, Siddhpur — 2018 – 2019

Class X — SMT S.B Doshi High School, Nandotra — 2016 – 2017

LANGUAGES

English (Proficient) • Hindi (Proficient) • Gujarati (Native)